

Lecture 21: Oct 10

Last time

- Internal midterm evaluation open

Today

- Extra office hour vote
- Moments

Moments

Example (Minimizing distance) The expected value of a random variable has another property, one that we can think of as relating to the interpretation of EX as a good guess at a value of X .

Suppose we measure the distance between a random variable X and a constant b by $(X - b)^2$. The closer b is to X , the smaller this quantity is. We can now determine the value of b that minimizes $E[(X - b)^2]$ and, hence, will provide us with a good predictor of X . (Note that it does no good to look for a value of b that minimizes $(X - b)^2$, since the answer would depend on X , making it a useless predictor of X .)

We could proceed with the minimization of $E(X - b)^2$ by using calculus, but there is a simpler method:

$$\begin{aligned} E(X - b)^2 &= E(X - EX + EX - b)^2 \\ &= E[(X - EX) + (EX - b)]^2 \\ &= E(X - EX)^2 + (EX - b)^2 + 2E[(X - EX)(EX - b)], \end{aligned}$$

where we have expanded the square. Note that $E[(X - EX)(EX - b)] = (EX - b)E(X - EX) = 0$, since $EX - b$ is constant and comes out of the expectation, $E(X - EX) = EX - EX = 0$. This means

$$E(X - b)^2 = E(X - EX)^2 + (EX - b)^2.$$

Such that $E(X - b)^2$ is minimized at $b = EX$. And $E(X - EX)^2$ is actually the variance of X ($Var X = E(X - EX)^2$).

The various moments of a distribution are an important class of expectations.

Definition For each integer n , the n th *moment* of X (or $F_X(x)$), μ'_n , is

$$\mu'_n = EX^n.$$

The n th *central moment* of X , μ_n , is

$$\mu_n = E(X - \mu)^n,$$

where $\mu = \mu'_1 = EX$.

Notes:

- $\mu'_0 = EX^0 = 1$
- μ'_1 is the *mean*, usually denoted by μ .
- $\mu_0 = E(X - \mu)^0 = 1$
- $\mu_1 = 0$
- $\mu_2 = E(X - EX)^2$ is the *variance*
- $\mu_3 = E(X - EX)^3$ is related to the *skewness*.
- $\mu_4 = E(X - EX)^4$ is related to the *kurtosis*.

Definition The *variance* of a random variable X is its second central moment, $\text{Var}(X) = E[(X - EX)^2]$. The positive square root of $\text{Var}(X)$ is the *standard deviation* of X .

The variance gives a measure of the degree of spread of a distribution around its mean. Figure 20.1 shows a plot of two samples, one sample draws 100 numbers from a normal distribution with mean 0 and variance 1, $N(0, 1)$. The other sample draws 100 numbers from a normal distribution with mean 0 and variance 100, $N(0, 100)$.

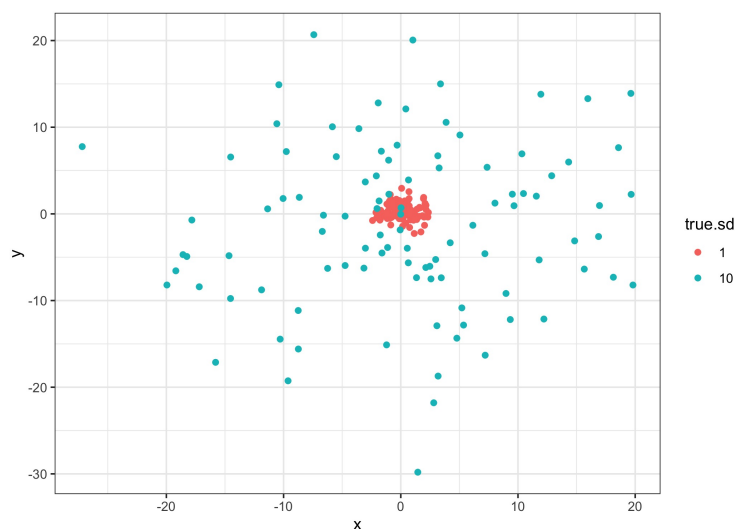


Figure 20.1: Figure 2.1.2. Two samples of 100 numbers drawn from $N(0, 1)$ and $N(0, 100)$.

Example (Exponential variance) Let X have the exponential(λ) distribution. We can calculate the variance of X now.

Solution:

$$\begin{aligned}\text{Var}(X) &= E(X - \lambda)^2 \\ &= \int_0^\infty (x - \lambda)^2 \frac{1}{\lambda} e^{-x/\lambda} dx\end{aligned}$$

Theorem If X is a random variable with finite variance, then for any constants a and b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Proof:

From the definition, we have

$$\begin{aligned}\text{Var}(aX + b) &= E[(aX + b) - E(aX + b)]^2 \\ &= E(aX - aEX)^2 \\ &= a^2 E(X - EX)^2 \\ &= a^2 \text{Var}(X).\end{aligned}$$

It is sometimes to use an alternative formula for the variance, given by

$$\text{Var}(X) = E(X^2) - (EX)^2,$$

which is easily established by

$$\begin{aligned}\text{Var}(X) &= E(X - EX)^2 = E[X^2 - 2XEX + (EX)^2] \\ &= EX^2 - 2(EX)^2 + (EX)^2 \\ &= EX^2 - (EX)^2.\end{aligned}$$

Example (Binomial variance) Let $X \sim \text{Binomial}(n, p)$, that is ,

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

What is the variance of X ?

Solutions:

Definition Let X be a random variable with cdf F_X . The *moment generating function (mgf)* of X (or F_X), denoted by $M_X(t)$, is

$$M_X(t) = Ee^{tX},$$

provided that the expectation exists for t in some neighborhood of 0. That is, there is an $h > 0$ such that, for all t in $-h < t < h$, Ee^{tX} exists. If the expectation does not exist in a neighborhood of 0, we say that the moment generating function does not exist.

More explicitly, we can write the mgf of X as

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx, \quad \text{if } X \text{ is continuous,}$$

or

$$M_X(t) = \sum_x e^{tx} \Pr(X = x), \quad \text{if } X \text{ is discrete.}$$

It is easy to see how the mgf generates moments as in the following theorem.

Theorem If X has mgf $M_X(t)$, then

$$EX^n = M_X^{(n)}(0),$$

where we define

$$M_X^{(n)}(0) = \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0}.$$

That is, the n^{th} moment is equal to the n^{th} derivative of $M_X(t)$ evaluated at $t = 0$.

Proof:

Example (Binomial mgf) Let $X \sim \text{Binomial}(n, p)$, then its mgf is

$$\begin{aligned} M_X(t) &= \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x} \\ &= [pe^t + (1-p)]^n. \end{aligned}$$

Theorem Let $F_X(x)$ and $F_Y(y)$ be two cdfs all of whose moments exist.

1. If X and Y have **bounded support**, then $F_X(u) = F_Y(u)$ for all u if and only if $EX^r = EY^r$ for all integers $r = 0, 1, 2, \dots$
2. If the moment generating functions exist and $M_X(t) = M_Y(t)$ for all t in some neighborhood of 0, then $F_X(u) = F_Y(u)$ for all u .

Theorem (Convergence of mgfs) Suppose $\{X_i, i = 1, 2, \dots\}$ is a sequence of random variables, each with mgf $M_{X_i}(t)$. Furthermore, suppose that

$$\lim_{i \rightarrow \infty} M_{X_i}(t) = M_X(t), \quad \text{for all } t \text{ in a neighborhood of } 0,$$

and $M_X(t)$ is an mgf. Then there is a unique cdf F_X whose moments are determined by $M_X(t)$ and, for all x where $F_X(x)$ is continuous, we have

$$\lim_{i \rightarrow \infty} F_{X_i}(x) = F_X(x).$$

That is, *convergence*, for $|t| < h$, of mgfs to an mgf implies *convergence* of cdfs.

Poisson approximation One approximation that is usually taught in elementary statistics courses is that binomial probabilities can be approximated by Poisson probabilities. It is taught that the Poisson approximation is valid “when n is large and np is small”, and rules of thumb are sometimes given.

The $Poisson(\lambda)$ pmf is given by

$$\Pr(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots,$$

where λ is a positive constant. The approximation states that if $X \sim Binomial(n, p)$ and $Y \sim Poisson(\lambda)$, with $\lambda = np$, then

$$\Pr(X = x) \approx \Pr(Y = x)$$

for large n and small np . We now show that the mgf converge, lending credence to this approximation. Recall that

$$M_X(t) = [pe^t + (1 - p)]^n.$$

For the $Poisson(\lambda)$ distribution, we can calculate (HW4, exercise 2.33)

$$M_Y(t) = e^{\lambda(e^t - 1)},$$

and if we define $p = \lambda/n$, then $M_X(t) = [1 + (e^t - 1)\lambda/n]^n$ such that $M_X(t) \rightarrow M_Y(t)$ as $n \rightarrow \infty$.

Theorem For any constant a and b , the mgf of the random variable $aX + b$ is given by

$$M_{aX+b} = e^{bt} M_X(at).$$

Proof:

Common Discrete Distribution

Why parametric models?

- *Parametric models* or *distribution families* have a specific form but can change according to a fixed number of parameters.
- The objective is to model a population. Parametric models are often appropriate in common situations with similar mechanisms.
- Parametric models have many known and useful properties and are easy to work with. When fitting a population, only a few parameters need to be estimated: *parametric inference*.
- Sometimes one does not want to make parametric assumptions and would rather work with non-parametric models. But non-parametric models can be infinite dimensional.
- In this course, we emphasize parametric models.

Discrete uniform X has the discrete uniform($1, N$) distribution if X is equally likely to be one of $\{1, 2, \dots, N\}$.

- Sample space: $\{1, 2, \dots, N\}$

- pmf:

$$f_X(x) = \frac{1}{N}, \quad x = 1, 2, \dots, N$$

- cdf:

$$F_X(x) = \Pr(X \leq x) = \begin{cases} 0 & x < 0 \\ \lfloor x \rfloor / N & 0 \leq x < N \\ 1 & N \leq x \end{cases}$$

- moments:

$$EX = \frac{N+1}{2}$$